Assignment 3 — Student Registration Analytics Console App

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Course: STY1991 — Software Engineering Concepts

Design Explanation

The StudentRegistration class contains StudentId, FullName, CourseCode, Credits, Mark and Status. StudentId is unique. The application starts with 10 registrations across CSC101, MAT101 and WEB101. Alex Smith and Diana Khan both have a mark of 85, resulting in a tie for the sorting tests. Three students are initially Waitlisted.

List<StudentRegistration> is the primary store. A synchronized Dictionary<string, StudentRegistration> provides lookup by student ID and duplicate checking. Both collections refer to the same objects, so a mark update is visible through either collection. Adding and deleting records changes both collections.

Queue<string» stores waitlist student IDs and processes them in first-in, first-out order. Processing changes valid waitlisted registrations to Enrolled; requests for deleted registrations are skipped. Stack<string> records successful Iditions, updates, deletions and waitlist enrollments. The history preview displays the latest three actions, newest first, without performing an undo.

Separate methods handle menu commands. Required text is trimmed and cannot be blank. TryParse validates numeric input; marks must be 0-100, and credits must be 1-6. A numbered choice restricts status to Enrolled, Waitlisted or Withdrawn. Student ID comparisons ignore case. A foreach loop displays records, while an index-based for loop finds and removes a matching list entry using RemoveAt.

LINQ Where filters enrolled students with marks of at least 70; OrderByDescending and ThenBy sort by mark and then name. Select projects names, course codes and marks. GroupBy and Count calculate per-course totals across all statuses. Average, Max and Min summarize enrolled marks only. Course searches ignore case. Empty results produce friendly messages, and an empty enrolled sequence is checked before statistics are calculated.

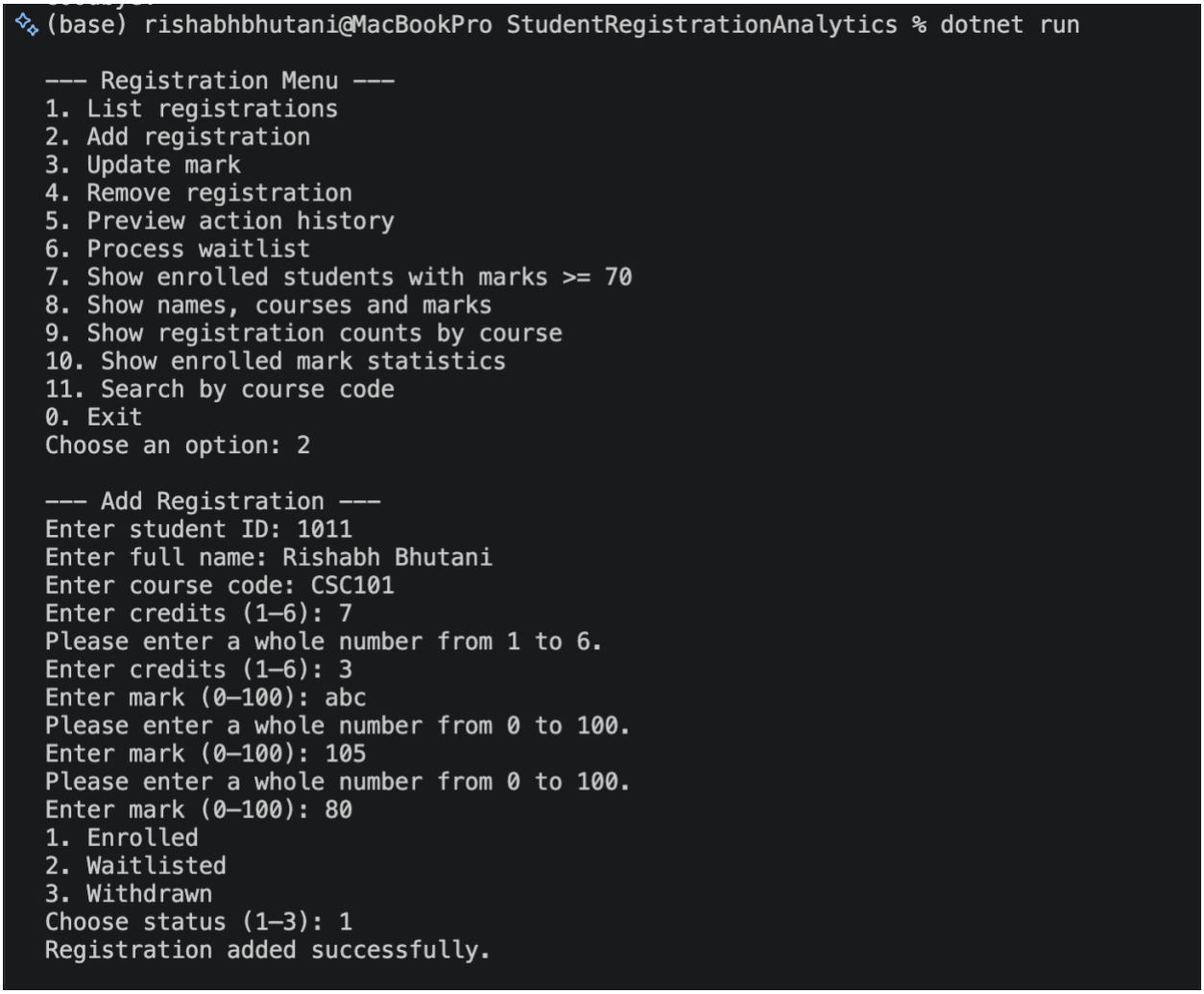
Test matrix

Test	Input	Expected or reported result	Status

T01	Add ID 1011 with credits 3, mark 80 and Enrolled status.	Accepted; record appears in list	Pass
T02	Attempt to add existing ID 1001	Duplicate rejected	Pass
T03	Enter credits 7, then 3; marks abc, 105, then 80	Invalid inputs rejected, valid values accepted	Pass
T04	Update ID 1011 from mark 80 to 95	List shows mark 95	Pass
T05	Delete ID 1011, then attempt a mark update	Deleted; dictionary lookup reports no registration	Pass
T06	Search CSC101, MAT101 and XYZ999 with seed data	3 matches, 4 matches and friendly no-match message	Pass
T07	Run high-mark report with seed data	Bella 92; Alex 85; Diana 85; Chris 78; Fatima 70	Pass
T08	Run enrolled statistics with seed data	6 enrolled; average 79.17; max 92; min 65	Pass
T09	Delete IDs 1001-1006; run options 10 and 7	Friendly empty-enrolled and no-high-mark messages.	Pass
T10	Delete all remaining records, run list, projection course counts and search	Friendly empty-result messages; no crash	Pass
T11	Process seeded waitlist, then process again	1007, 1009, 1010 processed in FIFO order; empty message	Pass
T12	Process queued requests after deleting their registrations	Deleted requests skipped safely	Pass

T13	Add, update and delete ID 1011; pre- view history	Delete, update, add shown newest first	Pass
T14	Run course counts with seed data	CSC101 3; MAT101 4; WEB101 3	Pass
T15	Run projected report with seed data	10 records with name, course code and mark only	Pass

Screenshots of validation and changes



Duplicate student IDs and invalid numeric inputs are rejected. A valid registration is accepted after correction.

```
(base) rishabhbhutani@MacBookPro StudentRegistrationAnalytics % dotnet run
Enter full name: Rishabh Bhutani
Enter course code: CSC101
Enter credits (1–6): 7
Please enter a whole number from 1 to 6.
Enter credits (1–6): 3
Enter mark (0–100): abc
Please enter a whole number from 0 to 100.
Enter mark (0–100): 105
Please enter a whole number from 0 to 100.
Enter mark (0–100): 80
1. Enrolled
2. Waitlisted
3. Withdrawn
Choose status (1–3): 1
Registration added successfully.

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 1

--- Student Registrations ---
1001 | Alex Smith | CSC101 | Credits: 3 | Mark: 85 | Enrolled
1002 | Bella Patel | CSC101 | Credits: 3 | Mark: 92 | Enrolled
1003 | Chris Lee | MAT101 | Credits: 4 | Mark: 78 | Enrolled
1004 | Diana Khan | WEB101 | Credits: 3 | Mark: 85 | Enrolled
1005 | Ethan Brown | MAT101 | Credits: 4 | Mark: 65 | Enrolled
1006 | Fatima Ali | WEB101 | Credits: 3 | Mark: 70 | Enrolled
1007 | George Wilson | CSC101 | Credits: 3 | Mark: 0 | Waitlisted
1008 | Hannah Chen | MAT101 | Credits: 4 | Mark: 40 | Withdrawn
1009 | Ishaan Sharma | WEB101 | Credits: 3 | Mark: 0 | Waitlisted
1010 | Julia Martin | MAT101 | Credits: 4 | Mark: 0 | Waitlisted
1011 | Rishabh Bhutani | CSC101 | Credits: 3 | Mark: 80 | Enrolled
```

The registration list includes student 1011 after a successful addition.

```
(base) rishabhbhutani@MacBookPro StudentRegistrationAnalytics % dotnet run
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 3

--- Update Mark ---
Enter student ID: 1011
Rishabh Bhutani — Current mark: 80
Enter new mark (0–100): 95
Mark updated successfully: 80 -> 95

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 1

--- Student Registrations ---
1001 | Alex Smith | CSC101 | Credits: 3 | Mark: 85 | Enrolled
1002 | Bella Patel | CSC101 | Credits: 3 | Mark: 92 | Enrolled
1003 | Chris Lee | MAT101 | Credits: 4 | Mark: 78 | Enrolled
1004 | Diana Khan | WEB101 | Credits: 3 | Mark: 85 | Enrolled
1005 | Ethan Brown | MAT101 | Credits: 4 | Mark: 65 | Enrolled
1006 | Fatima Ali | WEB101 | Credits: 3 | Mark: 70 | Enrolled
1007 | George Wilson | CSC101 | Credits: 3 | Mark: 0 | Waitlisted
1008 | Hannah Chen | MAT101 | Credits: 4 | Mark: 40 | Withdrawn
1009 | Ishaan Sharma | WEB101 | Credits: 3 | Mark: 0 | Waitlisted
1010 | Julia Martin | MAT101 | Credits: 4 | Mark: 0 | Waitlisted
1011 | Rishabh Bhutani | CSC101 | Credits: 3 | Mark: 95 | Enrolled
```

Updating student 1011 changes the mark from 80 to 95 in the displayed list

```
(base) rishabhbhutani@MacBookPro StudentRegistrationAnalytics % dotnet run

--- Remove Registration ---
Enter student ID: 1011
Registration removed successfully for Rishabh Bhutani.

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 3

--- Update Mark ---
Enter student ID: 1011
No registration found with that student ID.

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 5

--- Latest Three Successful Actions ---
Removed registration: 1011
Updated mark: 1011, 80 -> 95
Added registration: 1011
```

Deleting student 1011 removes the dictionary lookup. The stack preview lists deletion, update and addition in reverse chronological order

Screenshots of reports and waitlist processing

```
(base) rishabhbhutani@MacBookPro StudentRegistrationAnalytics % dotnet run
--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 7

--- Enrolled Students: Mark >= 70 ---
Bella Patel | CSC101 | Mark: 92
Alex Smith | CSC101 | Mark: 85
Diana Khan | WEB101 | Mark: 85
Chris Lee | MAT101 | Mark: 78
Fatima Ali | WEB101 | Mark: 70

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 10

--- Enrolled Mark Statistics ---
Enrolled students: 6
Average mark: 79.17
Highest mark: 92
Lowest mark: 65
```

The high-mark report sorts by descending mark, placing Alex before Diana when both have 85. Seed-data statistics show six enrolled students, an average of 79.17, a maximum of 92 and a minimum of 65

```
(base) rishabhbhutani@MacBookPro StudentRegistrationAnalytics % dotnet run

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 6

--- Process Waitlist ---
Processed 1007: George Wilson is now Enrolled.
Processed 1009: Ishaan Sharma is now Enrolled.
Processed 1010: Julia Martin is now Enrolled.
```

The queue processes IDs 1007, 1009 and 1010 in arrival order. A repeated request displays the empty-queue message

```
(base) rishabhbhutani@MacBookPro StudentRegistrationAnalytics % dotnet run

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 10

--- Enrolled Mark Statistics ---
No enrolled students available.

--- Registration Menu ---
1. List registrations
2. Add registration
3. Update mark
4. Remove registration
5. Preview action history
6. Process waitlist
7. Show enrolled students with marks >= 70
8. Show names, courses and marks
9. Show registration counts by course
10. Show enrolled mark statistics
11. Search by course code
0. Exit
Choose an option: 7

--- Enrolled Students: Mark >= 70 ---
No enrolled students have marks of 70 or higher.
```

After all enrolled registrations are deleted, the statistics and high-mark reports display friendly messages without calculating aggregates on an empty sequence